Test Documentation — Vehicle Washing Management System

Cover Page

Table 1

Document identification data

Field	Value
Project	Vehicle Washing Management System (Vehicle Washing / LavaRápido)
Deliverable	Test documentation (section 2.5.4)
Program	Software Analysis and Development Technologist (ADSO)
Group (ficha)	3145556
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Training center	Industria
Date	October 16, 2026

This document gathers the test plan, the inventory of automated tests in the repositories, their actual results when run on October 16, 2026 (including the end-to-end tests against the complete system), and the functional test cases, run on October 9, 2026, traced to the SRS v7 requirements (`lavarapido-docs/01_INFORME_DE_ESPECIFICACION_DE_REQUISITOS`). Its structure follows the ISO/IEC/IEEE 29119-3 standard for test documentation (International Organization for Standardization [ISO], 2021). No result in this document is assumed.

1. Test Plan

1.1 Scope

The web application, the mobile application, and the seven backend services on their `develop` branch are tested. Offensive security testing is out of scope; load is only checked through the RNF-001 response time measurement.

1.2 Test Types and Tools

Table 2

Test types, actual tool, and where each applies

Type	What it verifies	Tool	Where
Domain unit tests	Rules of entities and value objects (states, validations)	JUnit 5 (`spring-boot-starter-test`); xUnit 2	`src/test/.../domain` in Java; `tests/PaymentService.Domain.UnitTests`
Application unit tests	Use cases with in-memory repositories (test doubles)	JUnit 5 with fakes (`Fakes.java`, `InMemory*Repository`); xUnit with `InMemoryPorts.cs`	`application/usecase` and `PaymentService.Application.UnitTests`
Web integration (controller + security)	Routes, HTTP status codes, JWT, and roles with MockMvc	`spring-boot-starter-webmvc-test`, `spring-security-test`	`*ApiWebTest.java`, `ApiSecurityWebTest.java`, `AdminNotificationControllerTest.java`
Gateway route integration	Each public route reaches its service	Spring Boot Test	`GatewayRoutesTest.java`
Persistence integration	EF Core mapping against a real SQL Server	xUnit + `DatabaseFixture`	`PaymentService.Infrastructure.IntegrationTests` (requires a database)
Web UI unit tests	Angular components, guards, and services	Vitest 4 + jsdom (`@angular/build:unit-test`)	`Web/src/**/*.spec.ts`
Functional / acceptance	End-to-end flows per role against the deployed system	Manual, with Postman or Swagger UI for the API and a browser / Expo Go for the UI	Cases CP-01 to CP-33 (section 4)

Type	What it verifies	Tool	Where
Cross-service (end-to-end) and RNF	Booking with operator choice and automatic assignment, complete flow of the three roles, inspection report, access rules per role; 95th percentile response time (RNF-001); web screens at 360 px (RNF-008)	Node 24 (`node:test` and `fetch`) and Playwright Core with headless Edge	`lavarapido-infra/e2e`

Note. Tools documented in JUnit (n.d.), xUnit.net (n.d.), and Vitest (n.d.).

1.3 Environment

Table 3

Execution environment for the tests

Item	Version
Operating system	Windows 11 Home Single Language 10.0.26300 (October 15 and 16); 10.0.26200 (October 9)
Node.js / npm	24.14.0 / 11.9.0
JDK	Java 25 (the projects compile for Java 21) with each service's Maven Wrapper
.NET SDK	9.0.306; the project targets `net8.0` and the tests were run with `DOTNET_ROLL_FORWARD=Major`
Database and messaging	Not needed by the unit tests. The end-to-end tests of October 16 and the functional cases of October 9 used SQL Server 2022, RabbitMQ 4, and Mailpit on Docker 28.4 (Docker Desktop)
Services	For the functional cases, the 7 services on the machine (Spring Boot JARs and `dotnet run`); email was redirected to Mailpit so no real emails were sent

1.4 Criteria

- **Passed:** the actual result matches the expected one.
- **Failed:** the result differs; the error is recorded.
- **Blocked:** could not run because a dependency was missing (for example, the database).
- **Exit:** the release is accepted when the automated suites pass and the high-priority functional cases are passed.

2. Automated Test Inventory

The repositories have 542 test methods: 499 in the backend and 43 in the web. The mobile app has no automated tests.

Table 4

Tests per artifact (count of `@Test`/`@ParameterizedTest`, `[Fact]`/`[Theory]`/`[IntegrationFact]`, and `it()`)

Artifact	Files	Methods	What they cover
api-gateway	1	8	Routing of each `/api/v1/**` prefix to its service, and that `/internal/**` is not exposed by the gateway (`GatewayRoutesTest`)
security-service	17	131	Registration, sign-in with lockout after failed attempts, password recovery, profile and address, user administration, roles and permissions, preferences, value objects, password policy, JWT issuing and HTTP security by role (`ApiSecurityWebTest`)
customer-service	6	50	Customer account, license plates, vehicle CRUD, `security.user_registered` listener and web API
booking-service	6	40	Booking life cycle, availability, catalog, business data (`EstablishmentTest`), preferred operator in the event and web API
operations-service	11	54	Manual and automatic assignment, free operators, start window, execution, rating, operator metrics, team summary, operator profile, inspection report and web API
notification-service	12	106	Notification use cases, message factory per event, notices in the account's language, reminders, alerts for bookings without an operator, device tokens and admin controller

Artifact	Files	Methods	What they cover
payment-service (.NET)	9	110	Payment and receipt, reused reference, history with filters, promotions, loyalty points, payment policy, command and query services, and 7 persistence tests against SQL Server
Web (Angular)	37	43	Creation of components, guards and services (`should create` smoke tests), API error messages, and payment account selection
Mobile	0	0	No automated tests

> *Note.* Support files (`Fakes.java`, `InMemory*.java`, `InMemoryPorts.cs`, `DatabaseFixture.cs`) are not counted as tests. The 7 payment persistence tests skip themselves when the SQL Server connection variable is not set (`DatabaseFixture.cs:20-21`).

Backend tests are domain and application tests with in-memory doubles, plus controller tests with MockMvc. Most web tests check that each component is created; `api-error.spec.ts` and `payment.spec.ts` test real rules (the API error messages and the payment account selection).

3. Execution Results

On October 16, 2026, the eight automated suites were run: 639 tests, all passed. The 7 payment persistence tests were skipped because the test database was not set; on October 9 they had been run against an isolated database (`LavaRapidoPaymentIT`) created with the same changelogs, and they passed.

Table 5

Actual results per artifact (October 16, 2026)

Artifact	Command	Run	Passed	Failed	Skipped	Duration
Web (Angular)	`npx ng test --watch=false`	43	43	0	0	6.7 s
api-gateway	`./mvnw test`	34	34	0	0	9.9 s
security-service	`./mvnw test`	147	147	0	0	13.5 s
customer-service	`./mvnw test`	68	68	0	0	8.3 s
booking-service	`./mvnw test`	40	40	0	0	10.0 s
operations-service	`./mvnw test`	54	54	0	0	10.9 s
notification-service	`./mvnw test`	112	112	0	0	0.6 s
payment-service: domain	`dotnet test PaymentService.sln`	113	113	0	0	0.1 s
payment-service: application	(same command)	28	28	0	0	0.1 s
payment-service: integration	(same command, without `PAYMENT_INTEGRATION_DB`)	0	0	0	7	—
Total		**639**	**639**	**0**	**7**	

> *Note.* Sources: Vitest output, the `target/surefire-reports/*.xml` files of each Java service and the `dotnet test` output. Maven durations add up the time of the test classes, without compilation. There are more tests run than methods in Table 4 because parameterized tests (`@ParameterizedTest`, `[Theory]`) run once per data row.

Execution notes:

- The line `ERROR ... Could not send the password recovery email: SMTP down` in the security-service log is triggered on purpose by `EmailResetCodeNotifierTest`, which simulates an SMTP outage. It is not a failure.
- Vitest prints a build warning: `src/test-providers.ts` is not included in `tsconfig`. It does not affect the results.
- The run used JDK 25 and .NET SDK 9 on projects targeting Java 21 and .NET 8; for the final delivery it should be repeated with the project's exact versions.

3.1 End-to-End Tests

On October 16, 2026, `node --test "e2e/*.test.js"` was run from `lavarapido-infra`, with the containers of the *app* profile running and `E2E_OPERATOR_ID=2` (the test operator of the local database). All 19 tests passed in 7.3 s. Each test deletes what it creates and leaves the customer's points as they were.

Table 5a

End-to-end tests (October 16, 2026)

File	Tests	What it covers	Result
`access-rules.test.js`	7	Permissions per role and internal routes, profile address, payment history filters, tax data only for the administrator, operator profile, time outside opening hours (`reason: OUTSIDE_HOURS`), and the token of a nonexistent account	7 passed
`booking-flow.test.js`	5 (1 + 4 subtests)	Booking choosing an operator, automatic assignment through RabbitMQ, notices to the customer, day summary, and internal routes outside the gateway	5 passed
`full-flow.test.js`	6 (1 + 5 subtests)	Booking → reported payment → approval with points → service start and finish → rating, checking each notice	6 passed
`inspection-flow.test.js`	1	Inspection report with findings and photos, publication, notice to the customer, and public link	1 passed
Total	**19**		**19 passed**

4. Functional Test Cases

The 33 cases cover the requirements that were implemented or partial in SRS v4. They were run on October 9, 2026, between 15:40 and 15:45, against the full system through the API Gateway (the same calls the web and mobile clients make), using an automated script. All 33 passed. The requirements implemented afterwards (RF-004b, RF-008, RF-010, RF-015, RF-022, RF-036, RF-037 and RF-038) are covered by automated tests (section 5), and the booking flow with operator choice and automatic assignment is also run by `lavarapido-infra/e2e/booking-flow.test.js`.

Table 6

Functional test cases

ID	RF	Case	Precondition	Steps	Data	Expected result	Actual result	Status
CP-01	RF-001	Successful customer sign-up	Email not registered	Sign up and log in	ID, name, phone, new email, and valid password	Account created with the CLIENT role; the profile is created in customer-service	Sign-up 201; login with the CLIENT role; customer profile available (`GET /vehicles` 200)	Passed
CP-02	RF-001	Sign-up with a duplicate email	The email already exists	Sign up with the same email	Email from CP-01	"Email already registered" message; no account is created	409 `EMAIL_ALREADY_REGISTERED`; no account was created	Passed
CP-03	RF-002	Login by role	CLIENT, OPERATOR, and ADMIN users	Log in with each one	Valid credentials	Redirects to `/client`, `/operator`, or `/admin` by role	The three logins return a JWT with the ADMIN, OPERATOR, and CLIENT roles; the web redirects by that role (`authGuard`)	Passed
CP-04	RF-002	Wrong credentials	—	Log in with a wrong password, then with a nonexistent email	Two attempts	Same generic message in both; it does not reveal whether the email exists	Both: 401 `INVALID_CREDENTIALS` (same response)	Passed

ID	RF	Case	Precondition	Steps	Data	Expected result	Actual result	Status
CP-05	RF-003	Recover password	Registered user; Mailpit running	Request a code, read it in Mailpit, verify it, and set a new password	6-digit code	Password changed; login with the new one works	6-digit code received in Mailpit; verify 204, reset 204; login with the new password succeeds	Passed
CP-06	RF-003	Code locked after attempts	Code requested	Enter 5 wrong codes, then the right one	5 invalid codes	The right code is no longer accepted	5 attempts: 400; then the right code: 400 `INVALID_RESET_CODE`	Passed
CP-07	RF-004	Password change with a wrong current password	Logged in	Change the password with the wrong current one	Wrong current password	"Incorrect current password" error; nothing changes	400 `INCORRECT_CURRENT_PASSWORD`; login with the current password still works	Passed
CP-08	RF-005	Create a booking	Customer with an SUV	Choose vehicle, Basic service, a free date and time	SUV, Basic	Booking created; price COP 23,000 (seed price of Basic for SUV)	Booking 201 RES-060003 for 10/10/2026 08:00, status CONFIRMED; total COP 23,000	Passed
CP-09	RF-006	Time slot taken	Active bays of a slot booked	Book the same slot until it is full	Several customer vehicles	The booking is not saved; up to five alternative times on the same day are offered	Slot 08:30 full (BAY-02 and BAY-03 plus an earlier booking); the next one: 409 `SLOT_UNAVAILABLE` with alternatives 09:00, 09:30, 10:00, 10:30, and 11:00	Passed
CP-10	RF-007	Cancel a booking before it starts	Booking CONFIRMED	Cancel with a reason	Reason `CUSTOMER_REQUEST`	Booking is CANCELLED with the reason stored	200: status CANCELLED, reason `CUSTOMER_REQUEST`	Passed
CP-11	RF-007	Change a booking in progress	Booking IN_PROGRESS	Try to cancel	—	Action blocked with a message	Booking IN_PROGRESS; cancel: 409 `BOOKING_NOT_CHANGEABLE`	Passed
CP-12	RF-009	Assign an unavailable operator	Operator with an absence at the booking time	Check candidates and try to assign them; then assign an available one	Absent operator	The operator is not allowed; available operators are listed first	Available listed first; the absent one shows reason `OPERATOR_ABSENT`; assigning them: 409 `OPERATOR_ABSENT`; assigning the available one: 200	Passed
CP-13	RF-011	Operator sees their services	Booking assigned to the operator	Check assigned services	—	List with customer and vehicle data	RES-060003 shows vehicle Mazda CX-5, plate FPW-653, and service Basic	Passed
CP-14	RF-012b	Start and finish a service	Assigned service PENDING, 15 min before its time	Another operator tries to start; the assigned one starts and finishes	—	States IN_PROGRESS and COMPLETED; the other operator gets an error	Other operator: 404 `BOOKING_NOT_FOUND`; start 200 (IN_PROGRESS), finish 200 (COMPLETED); booking COMPLETED	Passed
CP-15	RF-012	Rate a service once	Booking COMPLETED	Rate 5 stars; try to rate again	5, comment	First rating saved; the second is rejected	First: 201 (5); second: 409 `ALREADY_RATED`	Passed
CP-16	RF-013	Payment methods	Payment accounts configured	Check methods (admin) and accounts (customer)	—	Cash, Nequi, Daviplata, and bank transfer; cash shows no QR	The admin sees the 4 methods; the API gives the customer only the active digital accounts with QR, and the UI always adds cash without QR (`payment.ts:44-57`)	Passed
CP-17	RF-013b	Upload a receipt	Booking awaiting payment	Pay with Nequi attaching image, amount, and reference	PNG < 10 MB	Payment created in review with its receipt	201: status IN_REVIEW, amount COP 24,000, reference REF-428646	Passed

ID	RF	Case	Precondition	Steps	Data	Expected result	Actual result	Status
CP-18	RF-014	Approve and reject payments	Payments in review	Approve one and reject another with a reason	Rejection reason	States APPROVED and REJECTED; the customer gets a notification	Approve 200 (APPROVED); reject 200 (REJECTED); the customer received `PAYMENT_CONFIRMED` and `PAYMENT_REJECTED`	Passed
CP-19	RF-016	Duplicate reference	Approved payment with reference R1	Upload another receipt with R1 on another booking	Reference R1	Receipt rejected	409 `TRANSACTION_REFERENCE_REUSED`	Passed
CP-20	RF-018	Booking notification	`MESSAGING_ENABLED=true`	Confirm a booking; open notifications; mark as read	—	The notification arrives; the unread counter goes down	9 notifications (`USER_WELCOME`, `BOOKING_CONFIRMED`, `BOOKING_CANCELLED`, `OPERATOR_ASSIGNED`); unread 9 → 8	Passed
CP-21	RF-020	Language saved on the account	Logged in	Switch to English; start a new session	`en`	The UI opens in English	Preference saved (200); the new session returns `language=en` and the web opened in English (test screenshot)	Passed
CP-22	RF-021	Schedule exception	—	Create a "closed" exception and check availability that day	10/20/2026	No times available that day	Exception 201; availability: `open=false`, no free times	Passed
CP-23	RF-024	Create and deactivate a promotion	Admin session	Create a 10% promotion and deactivate it	10% discount	Promotion visible, then inactive	Created 201; deactivated 200; status in the list: `paused`	Passed
CP-24	RF-026	Earn points	Service with 10 points configured	Approve the booking payment; check the balance	—	The balance grows by the service points	Balance 0 → 10 on payment approval	Passed
CP-25	RF-027	Deactivate one's own account	Customer session	Deactivate the account with the current password and try to log in	Current password	The session is revoked and login is rejected	Deactivation 204; new login: 403 `ACCOUNT_DISABLED`	Passed
CP-26	RF-028	Create an operator and restrict by role	Admin and customer sessions	Admin creates an operator; the customer calls `GET /admin/users`	Operator data	OPERATOR account created; the customer gets 403	Operator created 201; the customer gets 403	Passed
CP-27	RF-029	Edit the permission matrix	Admin session	Remove `EDIT_DATA` from OPERATOR; try to create a role	Permission `EDIT_DATA`	The matrix is saved; roles cannot be created	Matrix saved (200); create role: 405. The matrix was restored afterward	Passed
CP-28	RF-030	Register a vehicle with a duplicate plate	Customer with a registered vehicle	Register another vehicle with the same plate	Plate FPW653	Registration is rejected	First registration 201; second: 409 `PLATE_ALREADY_REGISTERED`	Passed
CP-29	RF-031	Price change without affecting bookings	Confirmed booking with Basic	Admin raises the Basic/SUV price; check the existing booking and create a new one	COP 23,000 → 24,000	The existing one keeps its price; the new one uses the new price	Earlier booking: COP 23,000; new booking: COP 24,000. The price was restored afterward	Passed

ID	RF	Case	Precondition	Steps	Data	Expected result	Actual result	Status
CP-30	RF-032	Bay under maintenance	BAY-04 under maintenance (seed)	Fill a slot and check which bays were used	—	The bay under maintenance is not used for new bookings	BAY-04 was not used; the slot was full with only the active bays	Passed
CP-31	RF-033	Disallowed status transition	Booking CANCELLED	Admin tries to change it to CONFIRMED	—	Rejected	409 `INVALID_STATUS_TRANSITION`	Passed
CP-32	RF-034	Deactivated payment account	Active Daviplata account	Admin deactivates it; the customer checks accounts	—	The account is no longer shown to the customer	Visible when created; once deactivated (200) it no longer appears to the customer	Passed
CP-33	RF-035	Refund an approved payment	Payment APPROVED	Admin refunds	—	Status REFUNDED; `payment.refunded` is published and the customer is notified	Refund 204; payment REFUNDED; `PAYMENT_REFUNDED` notification received; points reversed (10 → 0)	Passed

Note. Prices and codes in the test data come from the seed changelogs (`103-seed-catalog.sql`, `104-seed-booking-catalogs.sql`). Cases CP-25 to CP-33 cover requirements RF-027 to RF-035, new in SRS v4. New test accounts were used, because the demo account `operador@gmail.com` is deactivated in the local database. Afterward, the Basic price, the permission matrix, the promotion, and the schedule exception were restored. Evidence: execution log with the HTTP status and error code of each step.

5. Traceability Matrix

Of the 41 requirements in SRS v7, 30 have a functional case and 39 a related automated test; RF-032 and RF-034 only have a functional case.

Table 7

RF → status → endpoint / screen → test case

RF	Status (SRS v7)	Endpoint (prefix `/api/v1`)	Web screen	Functional case	Related automated test
RF-001	Implemented	`POST /auth/register`	`/auth/register`	CP-01, CP-02	`AccountRegistrationServiceTest`, `UserRegisteredListenerTest`
RF-002	Implemented	`POST /auth/login`	`/auth`	CP-03, CP-04	`AuthenticationServiceTest`, `JwtTokenIssuerTest`, `ApiSecurityWebTest`
RF-003	Implemented	`POST /auth/password/forgot`, `/verify`, `/reset`	`/auth/forgot-password`	CP-05, CP-06	`PasswordRecoveryServiceTest`, `PasswordResetTokenTest`
RF-004	Implemented	`PUT /users/me/password`	`/client/profile`	CP-07	`UserProfileAndAdministrationTest`, `PasswordPolicyTest`
RF-004b	Implemented	`POST /auth/login` (423 `ACCOUNT_LOCKED`)	`/auth`	—	`AuthenticationServiceTest` (lockout and window), `PasswordRecoveryServiceTest` (a reset lifts it), `ApiSecurityWebTest` (423)
RF-027	Implemented	`GET/PATCH /users/me`, `GET/PUT /users/me/address`, `PUT /users/me/email`, `POST /auth/logout`, `POST /users/me/deactivate`	`/client/profile`	CP-25	`UserProfileAndAdministrationTest`, `PersonAddressServiceTest`, `AuthenticationServiceTest`
RF-028	Implemented	`GET/POST /admin/users`, `PATCH /admin/users/{id}/status`	`/admin/management`, `/admin/operators`	CP-26	`UserProfileAndAdministrationTest`, `ApiSecurityWebTest`
RF-029	Implemented	`GET /admin/roles-permissions`, `PUT /admin/roles-permissions/{role}`	`/admin/management`	CP-27	`RolePermissionAdministrationServiceTest`
RF-030	Implemented	`/vehicles`, `GET /vehicle-types`, `GET /admin/vehicles`	`/client/vehicles`	CP-28	`CustomerVehicleServiceTest`, `LicensePlateTest`, `CustomerApiWebTest`
RF-031	Implemented	`GET /catalog/services`, `/admin/catalog/services`	`/admin/management`	CP-29	`CatalogServiceTest`

RF	Status (SRS v7)	Endpoint (prefix `/api/v1`)	Web screen	Functional case	Related automated test
RF-021	Implemented	`PUT /admin/schedule/business-hours`, `POST /admin/schedule/exceptions`	`/admin/schedule`	CP-22	`AvailabilityPolicyTest`
RF-032	Implemented	`/admin/bays`	`/admin/schedule`	CP-30	—
RF-037	Implemented	`GET/PUT /admin/establishment`, `GET /establishment`	`/admin/settings`	—	`EstablishmentTest`, `BookingApiWebTest`
RF-005	Implemented	`POST /bookings`, `GET /catalog/services`	`/client/reserve`	CP-08	`BookingServiceTest`, `BookingTest`
RF-006	Implemented	`GET /bookings/availability`	`/client/reserve`	CP-09	`AvailabilityPolicyTest`
RF-007	Implemented	`PUT /bookings/{id}`, `POST /bookings/{id}/cancel`	`/client/history`	CP-10, CP-11	`BookingTest`, `BookingApiWebTest`
RF-033	Implemented	`GET /admin/bookings`, `PATCH /admin/bookings/{id}/status`	`/admin/reservations`	CP-31	`BookingTest`, `BookingServiceTest`
RF-008	Implemented	`booking.confirmed` event, `GET /internal/v1/bookings`	`/admin/reservations`	—	`AutoAssignmentPolicyTest`, `OperationsServiceTest` (automatic assignment, deactivated account), `booking-flow.test.js`
RF-009	Implemented	`GET /admin/assignments/{id}/candidates`, `PUT /admin/assignments/{id}`	`/admin/reservations`	CP-12	`AssignmentPolicyTest`, `OperationsServiceTest`
RF-010	Implemented	`GET /operators/available`, `POST /bookings` (`preferredOperatorId`)	`/client/reserve`	—	`OperationsServiceTest` (free operators), `OperationsApiWebTest`, `booking-flow.test.js`
RF-011	Implemented	`GET /operator/services`	`/operator/assigned-services`	CP-13	`OperationsServiceTest`
RF-012b	Implemented	`POST /operator/services/{id}/start`, `/finish`	`/operator/assigned-services`	CP-14	`StartPolicyTest`, `ServiceExecutionTest`
RF-012	Implemented	`POST /ratings/{bookingId}`	`/client/history`	CP-15	`ServiceExecutionTest`
RF-022	Implemented	`PUT /admin/operators/{id}/availability`, `/absences`, `GET /admin/operators/summary`	`/admin/operators/:id`	—	`AssignmentPolicyTest`, `TeamSummaryCalculatorTest`
RF-036	Implemented	`/admin/inspections/{bookingId}/**`, `GET /public/inspections/{token}`	`/admin/reservations/:id/inspection`, `/report/:token`	—	`InspectionReportTest`, `OperationsApiWebTest`
RF-038	Implemented	`GET /admin/operators/profiles`, `PUT /admin/operators/{id}/profile`	`/admin/operators/:id`	—	`OperatorProfileServiceTest`
RF-034	Implemented	`/admin/payment-accounts`, `GET /payment-accounts`	`/admin/settings`	CP-32	—
RF-013	Implemented	`GET /payment-accounts`	`/client/payment`	CP-16	`PaymentQueryServiceTests`
RF-013b	Implemented	`POST /payments`	`/client/payment`	CP-17	`PaymentTests`, `PaymentCommandServiceTests`
RF-014	Implemented	`POST /admin/payments/{id}/approve`, `/reject`	`/admin/payments`	CP-18	`PaymentTests`, `PaymentCommandServiceTests`
RF-015	Implemented	`GET /payments/me?from=&to=&service=`	`/client/history`	—	`PaymentQueryServiceTests` (order and filters)
RF-016	Implemented	`POST /payments`	`/client/payment`	CP-19	`PaymentPolicyTests`, `PaymentCommandServiceTests` (reused reference)
RF-035	Implemented	`POST /admin/payments/{id}/refund`	`/admin/payments`	CP-33	`PaymentTests`, `PaymentCommandServiceTests`
RF-017	Implemented	`GET /admin/bookings/day-summary`, `GET /operators/my-bookings` and dashboard stores	`/client`, `/operator`, `/admin`	—	`BookingServiceTest` (day summary), web smoke tests (`dashboard.spec.ts`)
RF-018	Implemented	`/notifications/*` + RabbitMQ events	`/*/notifications`	CP-20	`NotificationUseCasesTest`, `NotificationFactoryTest`, `BookingReminderServiceTest`, `LocalizedNotificationsTest`, `UnassignedBookingAlertTest`, `StaffNotificationsTest`

RF	Status (SRS v7)	Endpoint (prefix `/api/v1`)	Web screen	Functional case	Related automated test
RF-019	Implemented	`PUT /users/me/preferences`	`/*/configuration`	Covered by CP-21	`UserPreferenceServiceTest`
RF-020	Implemented	`PUT /users/me/preferences`	`/*/configuration`	CP-21	`UserPreferenceServiceTest`, `UserPreferenceTest`
RF-023	Partial	`GET /admin/operators/{id}/metrics`, `GET /admin/operators/summary`, payment and booking stores	`/admin/reports`, `/admin/operators/:id`	—	`OperatorMetricsCalculatorTest`, `TeamSummaryCalculatorTest`
RF-024	Implemented	`/admin/promotions`	`/admin/management`	CP-23	`PromotionTests`
RF-025	Implemented	`POST /loyalty/redeem`	`/client/payment`	—	`CustomerPromotionServiceTests`
RF-026	Implemented	`GET /loyalty/balance`	`/client`	CP-24	`LoyaltyTransactionTests`

Note. The automated test column points to the file whose name and methods match the requirement; code coverage was not measured. `booking-flow.test.js` is in `lavarapido-infra/e2e` and needs the system running.

6. Findings and Open Items

Table 8

Testing gaps found

#	Gap	Recommendation
1	The mobile app has no automated tests	Add Jest with `jest-expo`, at least for validators (`shared/validators`) and services (`core/services`)
2	Most of the 43 web tests only check that the component is created	Test more real rules: `authGuard` by role, `authInterceptor` on 401, booking form
3	The 7 payment persistence tests need a migrated database and skip without it	Document in the README how to create the test database and set `PAYMENT_INTEGRATION_DB`
4	For RF-023 (reports) only the operator metrics and the team summary are tested; the PDF/Excel export has no test	Add a functional case for reports and the PDF/Excel export
5	RF-032 (bays) and RF-034 (payment accounts) have no dedicated automated tests: `ScheduleService` and `PaymentAccountService` have no test class	Add unit tests for those two services
6	The requirements implemented after October 9 (RF-004b, RF-008, RF-010, RF-015, RF-022, RF-036, RF-037, and RF-038) have automated tests but no functional case of their own; RF-008, RF-010, RF-015, RF-036, and RF-037 are already covered by the end-to-end tests	Add their CP cases and run them against the full system

Table 9

Defects found while running the system

#	Defect	Evidence	Severity
1	The administrator dashboard always says "¡Hola, Laura!" regardless of who logs in	`Web/src/app/features/admin/pages/dashboard/dashboard.ts:73` (`adminName = 'Laura'`); "Laura Méndez" is also hard-coded in `payment-review-modal.html:167`	Low
2	In the operator's "Assigned services" (web), the Customer column shows the license plate	`Web/src/app/features/operator/pages/assigned-services/assigned-services.ts:141`	Medium
3	On the customer home (mobile), "Assigned operator" is empty even though the booking has an operator	Booking RES-060003, assigned in CP-12; mobile screenshot	Medium
4	The brand name is inconsistent: "AquaWash" (mobile), "Car Wash" and "Lavado Vehicular" (web), "Express Car Wash" (establishment)	Web and mobile screenshots	Low

#	Defect	Evidence	Severity
5	The landing page advertises a "mobile wash service" and "wherever you are", but SRS v4 defines a single-location car wash	Landing screenshot; SRS v4, section 1	Low
6	If there are two accounts of the same method (two Nequi), the payment screen shows both selected at once	Customer Payments screenshot	Low

Note. Code review of October 16, 2026: the six defects are fixed. The dashboard greets with the session name and the payment review shows the reviewer (1); the Client column shows the customer name (2); the customer home shows the assigned operator on the web and on mobile (3); the brand is LavaRápido on the web and on mobile (4); the home page describes the service at the car wash (5); and the payment screen selects by account and shows the holder when two accounts share a method (6, `payment.spec.ts`).

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